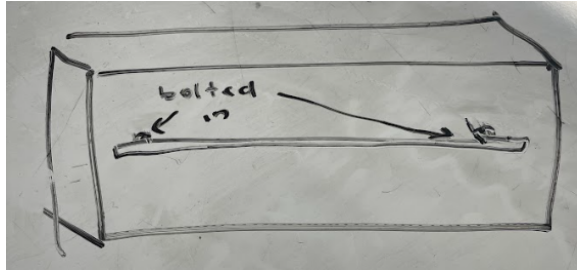
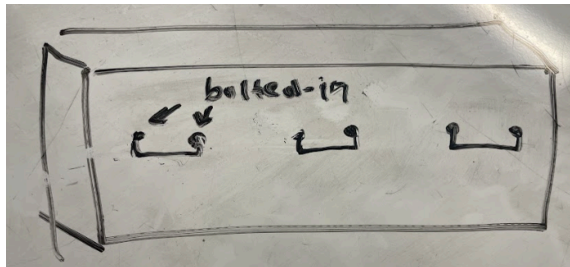
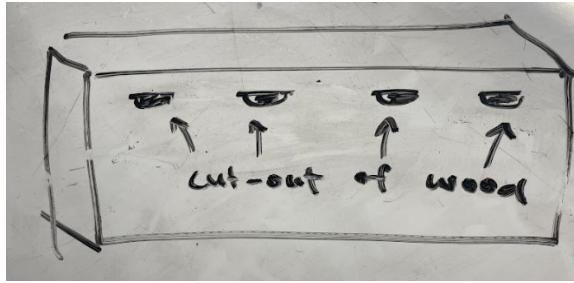


Komodo Dragon Element D

MATERIALS POSSIBLE DESIGN SOLUTIONS

We used two decision matrices to compare possible design solutions. We created a matrix for materials and a matrix for box handle designs and a matrix for materials. For the handles, we had five main design concepts:

Concepts	Picture
Static Long Bar - One long bar on each side of the box (total of six long bars needed). This would not be able to fold into the box, but would be very stable.	
Static Handles - Similar concept to the long bar but with handles that do not fold in. Still would be very stable.	
Folding Long Bar - Long bar on each side of each box. This can fold down to the sides as shown in the picture to make storage of the boxes easier.	

Folding Handles - Handles that are able to fold down to increase adjustability and decrease storage room needed. Similar concept to the folding long bar, but with handles.	
Divots - Indents in the frame of the design that would be large enough for hands to grip and hold the box up.	

Design Matrix Criteria

Choose appropriate weights for each criterion, listing each as a decimal	
Strength	0.3
Comfort	0.2
Foldability	0.2
Simplicity	0.2
Relative Material Cost	0.05
Repairability	0.05
Sum	1

All of our team members ranked the handle types based on the criteria.

Cole	Weight as	Static Long	Static	Folding Long	Folding	Divots
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	decimal	Bar	Handles	Bar	Handles	
Strength	0.3	8	9	7	8	10
Simplicity	0.2	10	10	6	6	8
Foldability	0.2	1	1	9	9	10
Comfort	0.2	5	8	5	8	1
Repairability	0.05	9	7	6	6	1
Relative Material						
Cost	0.05	6	9	5	8	10
	total score	6.35	7.3	6.65	7.7	7.35
Bennett	Weight as decimal	Static Long Bar	Static Handles	Folding Long Bar	Folding Handles	Divots
Strength	0.3	9	9	8	7	10
Simplicity	0.2	10	9	7	6	9
Foldability	0.2	1	1	10	10	2
Comfort	0.2	9	9	9	9	2
Repairability	0.05	10	8	7	5	2
Relative Material						
Cost	0.05	7	8	6	7	10
	total score	7.55	7.3	8.25	7.7	6.2
Kaitlin	Weight as decimal	Static Long Bar	Static Handles	Folding Long Bar	Folding Handles	Divots
Strength	0.3	9	9	7	7	10
Simplicity	0.2	9	9	6	6	9
Foldability	0.2	1	1	10	10	1
Comfort	0.2	10	9	10	9	1

Repairability	0.05	9	9	7	7	2
Relative Material						
Cost	0.05	7	8	5	6	10
	total score	7.5	7.35	7.9	7.75	5.8
Colt	Weight as decimal	Static Long Bar	Static Handles	Folding Long Bar	Folding Handles	Divots
Strength	0.3	8	8	7	7	9
Simplicity	0.2	9	9	5	5	9
Foldability	0.2	1	1	10	10	0
Comfort	0.2	9	5	6	5	2
Repairability	0.05	8	8	4	4	1
Relative Material						
Cost	0.05	8	8	8	8	9
	total score	7	6.2	6.9	6.7	5.4

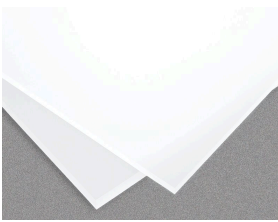
After our whole team gave a 1-10 rating for each of the designs based on the criteria, the best solution for handles was calculated.

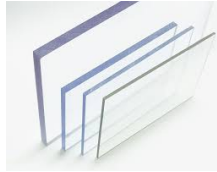
	Static Long	Static	Folding Long	Folding	
Results	Bar	Handles	Bar	Handles	Divots
Total Team					
Scores	28.4	28.15	29.7	29.85	24.75

With these results, we decided to ask the Zoo experts what they think would be preferable on a box for the dragons. They are leaning towards a folding design, and they also want handles on the ends of the boxes. We decided to make folding handles for the box.

POSSIBLE DESIGN MATERIALS

Similarly to the handles, we had to choose the main materials for the frame, floors, walls, and windows of the box. Five of the materials that we narrowed down to were:

Materials	Picture
Wood - Use regular wood as the frame for all three boxes. This would be strong but wouldn't last as long as some of the other materials.	
Aluminum - Use aluminum as the frame for the boxes. This would last longer than materials such as wood and the zookeepers plan to use this box until it breaks down.	
Polypropylene - Use this for the floors and walls of the three boxes because x-ray can scan through this.	

Polycarbonate - Use as windows so they can see the dragon while they are operating on it.	
Pressure-Treated Wood - Similar to regular wood, but lasts longer without rotting or extra damage.	

Strength	0.3
Animal Safety	0.3
Weight	0.15
Cost	0.1
Weather Resistance	0.1
Ease to Work With	0.05
Sum	1

As with the handles matrix, the results were calculated for the materials after we input our numbers.

	Weight as			Polypropylene	Polycarbonate	Pressure-Treated
Cole	decimal	Wood	Aluminum			Wood*

Weather						
Resistance	0.1	4	8	9	10	6
Cost	0.1	10	5	6	6	8
Strength	0.3	5	9	7	7	7
Ease to Work						
With	0.05	8	6	7	7	2
Weight	0.15	6	7	9	9	6
Animal Safety	0.3	7	9	8	9	3
	total score	6.3	8.05	7.7	8.1	5.4
Bennett	Weight as decimal	Wood	Aluminum	Polypropylene	Polycarbonate	Pressure-Treated Wood*
Weather						
Resistance	0.1	5	7	9	9	7
Cost	0.1	10	6	7	6	7
Strength	0.3	6	8	6	7	6
Ease to Work						
With	0.05	9	7	6	6	9
Weight	0.15	5	7	9	9	5
Animal Safety	0.3	6	8	8	8	6
	total score	6.3	7.5	7.45	7.65	6.2
Kaitlin	Weight as decimal	Wood	Aluminum	Polypropylene	Polycarbonate	Pressure-Treated Wood*
Weather						
Resistance	0.1	4	8	9	9	7
Cost	0.1	10	5	6	6	6

Strength	0.3	7	9	7	7	7
Ease to Work						
With	0.05	10	5	5	5	7
Weight	0.15	4	8	8	8	4
Animal Safety	0.3	4	7	9	9	5
	total score	5.8	7.55	7.75	7.75	5.85
Colt	Weight as decimal	Wood	Aluminum	Polypropylene	Polycarbonate	Pressure-Treated Wood*
Weather						
Resistance	0.1	5	9	9	9	9
Cost	0.1	10	3	6	6	5
Strength	0.3	7	9	4	4	6
Ease to Work						
With	0.05	9	4	5	5	9
Weight	0.15	2	9	8	8	2
Animal Safety	0.3	5	5	9	9	5
	total score	5.85	6.95	6.85	6.85	5.45

Results	Wood	Aluminum	Polypropylene	Polycarbonate	Pressure-Treated Wood*
Total Team Scores	24.25	30.05	29.75	30.35	22.9

Aluminum, wood, and pressure-treated wood were our three options for the frame of the boxes. The results from this design matrix and inclinations from the zoo experts that metal would be preferable, we decided that aluminum for the frame would be the best option. Some other materials that we decided to use were polypropylene and polycarbonate.

The aluminum frame is preferable to wood for many reasons. The zookeepers explained that wood would be harder to clean. If, for example, the weather rain or mud got into the wood, then some of the water or particles might seep into it and start to deteriorate the wood. The wood will also start to decay due to weather conditions and time. An aluminum frame would be much easier to clean and would withstand the weather much better.

TIMELINE: Plan of action

